Solution:

Diameter, $d = 7\ cm$, therefore $r = \frac{7}{2}\ cm$.

Area of the shaded region = Area of the square – 4 × Area of the circular quadrant

$$= a^2 - 4 \times \frac{1}{4}\pi r^2$$

$$= (7 \times 7) - \left(4 \times \frac{1}{4} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2}\right)$$

$= 49 - 38.5 = 10.5\ sq.cm.$ (approximately)

Exercise 2.1

1. **Fill in the blanks:**

   (i) The ratio between the circumference and diameter of any circle is _____.

   (ii) A line segment which joins any two points on a circle is a _____.

   (iii) The longest chord of a circle is _____.

   (iv) The radius of a circle of diameter 24 *cm* is _____.

   (v) A part of circumference of a circle is called as _____.

2. **Match the following:**

   (i) Area of a circle - (a) $\frac{1}{4}\pi r^2$

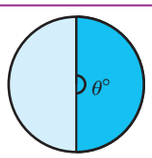
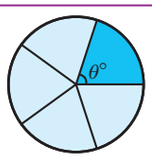
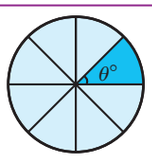
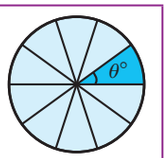
   (ii) Circumference of a circle - (b) $(\pi + 2)r$

   (iii) Area of the sector of a circle - (c) πr^2

   (iv) Circumference of a semicircle - (d) $2\pi r$

   (v) Area of a quadrant of a circle - (e) $\frac{\theta^\circ}{360^\circ} \times \pi r^2$

3. Find the central angle of the shaded sectors (each circle is divided into equal sectors).

Sectors				
Central angle of each sector (θ°)				

4. For the sectors with given measures, find the length of the arc, area and perimeter. (π=3.14)

   (i) central angle 45°, $r = 16\ cm$ (ii) central angle 120°, d =12.6 *cm*

5. From the measures given below, find the area of the sectors.

   (i) length of the arc = 48 *m*, $r = 10\ m$ (ii) length of the arc = 50 *cm*, r = 13.5 *cm*

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